



CENTRAL UNIVERSITY OF SOUTH BIHAR

Department of Computer Science

End Semester Examination Dec-2025

Programme: M.Sc (AI)

Course Title: Social Network Analytics

Maximum points: 70

Enrollment No.: CUSB7402332028

Course code: CAI91DE02204

Duration: 3:00 Hrs

Semester - 3rd

Instruction: Attempt questions in all the Sections as directed.

Section A

*Answer all questions below a maximum of 50 words, each carrying a maximum of 1 point.
(10 X 1 = 10)*

1. How have OSNs transformed our lives? Give any two reasons.
2. Define a network.
3. What are the advantages of studying social networks?
4. How can Social Network Analytics help in epidemiology?
5. Give any two characteristics of random networks.
6. List the key components of network topology.
7. Define preferential attachment property of social networks.
8. List the limitations of Watts-Strogatz Model.
9. Define anonymity within the context of Online Social Networks.
10. Define the span of a local bridge with an example.

Section B

*Answer any 5 questions below a maximum of 300 words, each carrying a maximum of 6 points.
(5 X 6 = 30)*

1. Explain Client-Server based architecture of OSNs.
2. Discuss the common properties exhibited by various Social Networks.
3. Define and compute closeness centrality while taking a small graph of your choice.
4. Compare scale free and non-scale free graph models.
5. Discuss Mark Granovetter's experiment. What are its implications? [4+2]
6. What is the Random-surfer model in the context of Page rank?

Section C

Answer any 2 questions below a maximum of 1000 words, each carrying a maximum of 15 points. *(2 X 15 = 30)*

1. Answer the following questions.

- a. Discuss various factors that guide the evolution of an online social network. [7]
- b. Define and explain the key factors that trigger a cascade behaviour. What are the factors that stop cascade from happening further? [4+4]

2. Answer the following questions:

- a. Define *Preferential attachment*. Explain Barabasi-Albert (BA) Model. [2+6]
- b. What do you understand about Power Law Distribution? Discuss its characteristics. [3+4]

3. Answer the following questions:

- a. Discuss Newman-Girvan Algorithm with a suitable example. Also list applications of detecting communities in OSNs. [5+3]
- b. Define privacy. How do OSNs create risk to users' privacy? [2+5]